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Systematic Review/Meta-analysis

Patient- and Family-Centered Care Recommendations in Cardiology Guidelines: An AI-Driven Systematic Review

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ABSTRACT

Background: Patient- and family-centred care (PFCC) is recognized as a critical component of cardiovascular care, but its integration into cardiology society guidelines has not been described. The objective of this study is to review PFCC language use and recommendations within major cardiology society guidelines.

Methods: We conducted a systematic review of guidelines and statements from the American College of Cardiology (ACC), the American Heart Association (AHA), the Canadian Cardiovascular Society (CCS), and the European Society of Cardiology (ESC) for the period 2013-2023. PFCC-related key terms were identified using an artificial intelligence–based natural language processing algorithm, and recommendations were categorized into 8 dimensions of PFCC. The inclusion of PFCC recommendations across societies and trends over time were examined.

Results: A total of 260 guidelines and statements were analyzed. The most frequent PFCC dimensions overall were Health Transitions (23.5 per 100 pages), Shared Decision-Making (11.1 per 100 pages), and Care Access (9.9 per 100 pages). The least commonly identified dimensions across all journals were Care Coordination (6.5 per 100 pages), Emotional Support (4.0 per 100 pages), and Familial Support (1.0 per 100 pages). The CCS, ACC, and AHA had more recommendations using PFCC key terms than the ESC per 100 pages (17.3, 12.0, 10.3 vs 4.6, respectively, $P < 0.01$). PFCC language usage increased markedly over the 10-year period for the ACC, AHA, and ESC, but it decreased for the CCS (all $P < 0.05$).

RÉSUMÉ

Contexte : Si les soins centrés sur le patient et la famille sont reconnus comme étant un élément essentiel des soins cardiovasculaires, leur intégration dans les lignes directrices des sociétés de cardiologie n'a pas été décrite. L'objectif de cette étude consistait à examiner l'utilisation de termes liés aux soins centrés sur le patient et la famille et les recommandations dans les lignes directrices des principales sociétés de cardiologie.

Méthodologie : Nous avons procédé à un examen systématique des lignes directrices et des énoncés de l'American College of Cardiology (ACC), de l'American Heart Association (AHA), de la Société cardiovasculaire du Canada (SCC) et de la Société européenne de cardiologie (SEC), de 2013 à 2023. Nous avons recensé les termes clés liés aux soins centrés sur le patient et la famille à l'aide d'un algorithme de traitement du langage naturel utilisant l'IA, et nous avons classé les recommandations en fonction de huit dimensions. Nous avons examiné l'inclusion des soins centrés sur le patient et la famille aux recommandations des sociétés et dégagé des tendances au fil du temps.

Résultats : Nous avons analysé 260 lignes directrices et énoncés. D'une manière générale, les dimensions liées aux soins centrés sur le patient et la famille les plus fréquentes ont été les suivantes : les transitions en santé (23,5/100 pages), la prise de décision partagée (11,1/100 pages) et l'accès aux soins (9,9/100 pages). Les dimensions les moins fréquentes dans toutes les revues ont été les suivantes : la coordination des soins (6,5/100 pages), le soutien

Keywords: patient- and family-centered care; cardiology guidelines; shared decision-making; health transitions; clinical recommendations; systematic review; healthcare communication

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Conclusions: PFCC language and recommendations are being included increasingly in cardiology society guidelines. Differences exist in PFCC language use across these societies. Future research is needed to evaluate the impact of these guideline recommendations on clinical practice.

émotionnel (4,0/100 pages) et le soutien familial (1,0/100 pages). La SCC, l'ACC et l'AHA comptaient plus de recommandations utilisant des termes clés liés aux soins centrés sur le patient et la famille que la SEC, par 100 pages (17,3, 12,0, 10,3 *vs* 4,6, respectivement; $p <$ 0,01). L'utilisation de termes liés aux soins centrés sur le patient et la famille a fortement augmenté en l'espace de 10 ans dans le cas de l'ACC, de l'AHA et de la SEC; elle a diminué dans le cas de la SCC ($p <$ 0,05 dans tous les cas).

Conclusions : Les termes et les recommandations liés aux soins centrés sur le patient et la famille sont de plus en plus présents dans les lignes directrices des sociétés de cardiologie. Il existe des différences dans l'utilisation de ces termes d'une société à l'autre. D'autres recherches s'imposent pour évaluer les répercussions de ces recommandations dans la pratique clinique.

Patient- and family-centred care (PFCC) has emerged as a key component of contemporary healthcare delivery.[1] PFCC involves a paradigm shift from a traditional disease-centred focus to an approach of collaboration, participation, and information sharing in a respectful manner with patients and their families.[2] A growing body of literature suggests that PFCC strategies can improve patient experiences and outcomes, including medical goal achievement, increased care satisfaction, and improved mental health,[3] while also providing systemic benefits such as reduced incidence of hospital readmissions and improved resource utilization.[3-6]

PFCC is recognized increasingly as an integral goal of cardiovascular care. Recent American Heart Association (AHA) Scientific Statements have highlighted the importance of integrating patient values and preferences into cardiovascular care decisions, and the role of engaging family members in care delivery to improve patient-oriented and clinical outcomes.[7,8] These strategies have been associated with improved adherence to prescribed therapies, care satisfaction, and quality of life (QOL), particularly in cardiovascular medicine, an arena in which patients often face complex, long-term management challenges.[9,10] Given the critical role of clinical guidelines in shaping practice, ensuring that PFCC principles are well represented in guideline recommendations is essential for fostering care that is more patient-centred.

The PFCC dimensions used in this study—Values and Preferences, Information and Education, Care Access, Emotional Support, Familial Support, Health Transition, Care Coordination, and Shared Decision-Making— encompass essential aspects of PFCC and are adapted from established frameworks.[11,12] However, the degree to which cardiovascular guidelines, which are pivotal in directing care for high-risk populations, incorporate these PFCC principles remains unclear.

Professional society guidelines play an integral role in defining practice standards and informing clinical practice. Including PFCC language and recommendations in these guidelines could increase awareness of patient-centred principles, with further efforts needed to disseminate and implement these recommendations in clinical practice. An understanding of how cardiology professional societies incorporate PFCC language and recommendations into their guidelines is needed. This study seeks to address this need by systematically analyzing the use of PFCC language and the emphasis on PFCC in major international cardiovascular guidelines, providing a foundation for further efforts to enhance the patient-centredness of clinical care.

The objective of this study is to review major American, Canadian, and European cardiovascular society guidelines for PFCC language use and care recommendations. This study hypothesizes that the frequency of PFCC language and recommendations in cardiology guidelines have increased over time, reflecting a shift toward more PFCC. By identifying these trends, our findings provide foundational insights for healthcare providers, policymakers, and guideline developers to use to explore strategies for enhancing the integration of PFCC principles into clinical care.

Methods

Eligibility criteria

We performed a systematic review of major American, Canadian, and European cardiovascular professional society guidelines published between January 2013 and September 2023. The timeframe of 2013—2023 was selected to enable the analysis of trends in PFCC language and recommendations over the past decade since the publication of the American College of Cardiology (ACC) 2012 Health Policy Statement on Patient-Centered Care in Cardiovascular Medicine.[13] All guidelines and statements published and posted on the ACC's Web site during this period were included; no other exclusion criteria were used. Eligible articles were extracted from the professional society Web sites. The search was conducted in September 2023. For articles that had more than one format (ie, an executive summary and a longer document), the longer document was included. This study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Supplemental Table S1).

Information sources

We reviewed both pediatric and adult guidelines from the following cardiovascular professional societies: the ACC, the

AHA, the Canadian Cardiovascular Society (CCS), and the European Society of Cardiology (ESC), given their international reach and influence, as evidenced by the high rate of citations of their guidelines.

Search strategy and data extraction

Key terms relevant to PFCC in cardiology were identified, based on standardized frameworks (eg, Institute of Medicine, Picker Institute), to provide uniform definitions across dimensions, from the AHA statement on patient-centred cardiovascular care (Supplemental Fig. S1). The PFCC key terms were derived from the AHA statement, which serves as a key PFCC-oriented document in cardiovascular medicine and synthesizes the most recent and comprehensive PFCC research. The key terms were extracted from the guideline and selected based on the concepts emphasized within it, reflecting universal PFCC principles applicable across all contexts. To ensure objectivity, the terms were validated by independent reviewers for contextual relevance to PFCC. An artificial intelligence (AI) language model was used solely to assist in generating an initial list of potential PFCC-related key terms, based on semantic analysis of existing literature. This list was reviewed and validated by 3 human reviewers to ensure the relevance and accuracy of the terms. The AI model was not involved in assigning key terms to PFCC dimensions, assessing contextual relevance, or performing any quantitative analyses. All final decisions regarding key terms and their inclusion in the analysis were made by human reviewers. The following key terms were included: beliefs, care coordination, caregiver involvement, communication, dignity, education, family-centred care, family communication, family involvement, family support, health literacy, home-based care, information, needs, patient-centred, patient-centred care, patient education, patient engagement, personal support, preference, psychosocial support, QOL, respect, shared decision-making (SDM), support network, surrogate decision-making, and values. Variant spellings of each term were included (eg, “patient-centered” and “patient centred”).

We used the Hugging Face BERT (for Bidirectional Encoder Representations from Transformers) language model[14] as a basis for our algorithm to generate semantically related key terms, which resulted in 240 semantically related key terms to be used to search the texts. Key terms identified by the AI model were reviewed and validated by 3 human reviewers (S.A.B., N.S.N., C.G.) to ensure their contextual relevance to PFCC principles. We included 204 semantically related key terms in the search. The articles were converted from .pdf to .txt format using a high-performance Python library data extractor to permit accurate and efficient key-term detection. Using the .txt files, our automated model identified all sentences containing the generated key terms.

This study required analyzing every line of text in 260 guidelines and statements published over the past decade. Given the scale and complexity of the data, a traditional manual approach would have been impractical. The use of an AI-based methodology enabled systematic, comprehensive, and reproducible identification of PFCC-related terms, providing a level of detail and accuracy unattainable through manual abstraction.

Once the key terms were finalized, all phrases containing any of the 204 identified key terms were extracted systematically from the text. For each extracted text string, at least 2 of 3 reviewers (S.A.B., N.S.N., C.G.) reviewed and determined whether the key term was used within the context of PFCC or in another context. Text strings that were unrelated to PFCC (eg, operational or administrative language) were excluded. Disagreements were resolved through consensus with a third reviewer. In addition to identifying whether the key term was used in a relevant context, we categorized the extracted text as either a specific recommendation or supporting text, herein referred to as “in-text.”

Categorization

We categorized recommendations and supporting text into the following dimensions of PFCC: (i) Respect for the Patient’s Values, Preferences, and Expressed Needs; (ii) Information and Education; (iii) Access to Care; (iv) Emotional Support to Relieve Fear and Anxiety; (v) Involvement of Family and Friends; (vi) Continuity and Secure Transition Between Healthcare Settings; (vii) Coordination of Care; and (viii) Shared Decision-Making. These 8 dimensions were adapted from established frameworks in patient-centred and family-centred care, particularly those outlined in the Institute of Medicine’s "Crossing the Quality Chasm" framework and the Picker Institute’s "Through the Patient’s Eyes" framework (Supplemental Appendix S1). These frameworks have been applied widely in healthcare contexts, to promote comprehensive patient care that considers both clinical outcomes and patient experiences.[11,12]

Key terms were manually assigned to the appropriate PFCC dimension (Supplemental Appendix S1). We then used an AI language model to verify these classifications. The AI language model was developed using the cloud-based JavaScript platform Apps Script, (Google, Mountain View, CA) which analyzed the full text of each guideline and attributed it to a specific PFCC dimension using the manually assigned key terms. The verification process ensured consistency and reduced the level of human bias in the categorization process.

Data standardization and analysis

To account for the varying lengths of guidelines and statements, we standardized the results by reporting key-term frequencies and recommendations per 100 pages. This metric was calculated across the entire length of each guideline or statement, and was not limited to any specific subset of pages. The choice of 100 pages as a denominator allows for meaningful comparisons across documents of different lengths and structures, ensuring consistency in reporting. The total number of pages for each document was calculated, excluding the reference section.

In the analysis, we examined each society individually, and joint statements issued by multiple societies (eg, AHA and ACC) were attributed to all contributing societies, treating them as unique publications for society-specific analyses. This approach ensured fair representation of each society’s contributions. Comparisons between societies were based on these attributions, to identify trends and differences.

For aggregate totals across all journals, joint statements were included only once, to avoid double-counting, ensuring

Table 1. Frequency of in-text occurrences of top 10 patient- and family-centred care (PFCC) terms across articles

Key term	*ACC*	*AHA*	*ESC*	*CCS*	*Average*
Shared decision-making	15.8	14.2	3.4	5.7	9.9
Values	4.9	5.0	1.5	5.7	7.8
Quality of life	8.1	8.9	4.8	8.7	7.1
Education	5.8	6.0	4.3	7.1	5.5
Multidisciplinary	4.7	4.3	1.6	11.3	4.0
Needs	3.0	4.2	2.0	3.5	3.0
Patient-centred	1.0	5.6	1.3	1.6	3.0
Communication	2.4	2.8	1.8	1.9	2.3
Patient preference	2.1	2.0	0.9	1.9	1.6
Quality of care	1.8	2.2	0.5	1.0	1.4

This table reports the frequency of the top 10 PFCC-related terms identified in cardiology guidelines and statements from the American College of Cardiology (ACC), the American Heart Association (AHA), the European Society of Cardiology (ESC), and the Canadian Cardiovascular Society (CCS). Terms are quantified as occurrences per 100 pages to standardize comparisons across varying document lengths. The average across journals provides an average perspective on PFCC language usage and helps identify collective trends among the societies.

Table 2. Frequency of occurrence of patient- and family-centred care (PFCC) dimensions across the societies

PFCC dimension	*ACC*	*AHA*	*CCS*	*ESC*	*Mean*
Health Transition	10.3	13.1	58.2	5.0	17.6
Values and Preferences	14.6	14.1	7.8	3.2	9.4
Shared Decision-Making	15.2	13.0	12.5	2.5	9.1
Information and Education	9.2	7.9	8.3	7.4	8.1
Care Access	8.1	10.0	9.7	4.9	7.7
Care Coordination	5.6	4.2	12.5	2.1	5.3
Emotional Support	5.1	6.0	3.5	2.8	4.2
Familial Support	1.7	1.1	0.6	0.2	0.9

This table displays the frequency of occurrences of various PFCC dimensions within cardiology guidelines from the American College of Cardiology (ACC), the American Heart Association (AHA), the European Society of Cardiology (ESC), and the Canadian Cardiovascular Society (CCS). The frequencies reported are per 100 pages, allowing for standardized comparison across guideline documents. The color scale represents relative prevalence: green indicates the highest frequencies, yellow indicates moderate frequencies, and red indicates the lowest frequencies of PFCC dimensions.

that the overall findings accurately reflected unique contributions. This approach allowed for both society-specific insights and an overarching view of PFCC-related language usage and trends. Although comparisons between societies provide valuable insights, they should be interpreted with the understanding that joint statements represent shared contributions.

Statistical analyses

The number of PFCC recommendations per 100 pages and per article, across the 4 societies, were compared using analysis of variance (ANOVA); a *post hoc* test was then used to determine which pairs of societies differ from one another significantly. Temporal trends in the use of PFCC dimensions were evaluated using ANOVA to assess for changes in the frequency of dimensions over time. ANOVA was also used to compare the rate of change, measured through linear regression, of PFCC dimensions across societies over the decade-long timeframe, to identify differences in trends. PFCC phrase counts within specific clinical contexts, such as across different cardiovascular disease topics, were compared relative to one another. All statistical tests were conducted with a significance threshold of $P < 0.05$. Analyses were performed using SPSS 27.0 software (IBM, Armonk, NY).

Results

Article inclusion

A total of 260 articles were included in the analysis (AHA, n = 58; ACC, n = 67; CCS, n = 46; ESC, n = 89; Supplemental Table S2; see Supplemental Appendix S1). In all, 146 guidelines and 114 statements were included.

In-text usage of PFCC terms across guidelines

The most frequently mentioned terms used in-text across journals were in the SDM dimension (9.9 occurrences per 100 pages), the Values and Preferences dimension (7.8 per 100 pages), and the QOL dimension (7.1 per 100 pages; Table 1; see Supplemental Fig. S2).

The most identified dimensions across all journals were Health Transition (17.6 per 100 pages), Values and Preferences (9.4 per 100 pages), and SDM (9.1 per 100 pages; Table 2). The least commonly identified dimensions across all journals were Care Coordination (5.3 per 100 pages), Emotional Support (4.2 per 100 pages), and Familial Support (0.9 per 100 pages). The most common dimensions differed by society, as follows: SDM and Values and Preferences, for the ACC and AHA ($P = 0.63$ between these 2 societies); Information and Education, and Health Transition for the ESC ($P = 0.005$ between the ESC and both the ACC and the AHA), and Health Transition and Care Coordination for CCS ($P = 0.006$ between the CCS and both the ACC and the AHA; Table 2). Familial Support was the least emphasized dimension across all societies.

Recommendations containing PFCC terms

A total of 501 recommendations included at least one PFCC term. Per 100 pages, the CCS (mean = 17.3), ACC (mean = 12.0) and AHA (mean = 10.3) had more recommendations using PFCC key terms than did the ESC (mean = 4.6, $P < 0.001$; $P < 0.01$; $P = 0.1$). Identified recommendations were classified by dimension and summarized (Supplemental Appendix S2; Supplemental Fig. S3). Per article, the ACC (mean = 4.4) and AHA (mean = 4.8) had more recommendations that included PFCC terms than did the CCS (mean = 2.2; $P = 0.07$; $P = 0.03$) and the ESC (mean = 1.5; $P < 0.001$). The AHA had the highest number of recommendations per single article—36.[15]

The recommendations containing a PFCC key term were categorized by PFCC dimension. The most frequently addressed categories were Values and Preferences (1.8 per 100 pages), SDM (1.4 per 100 pages), and Information and Education (1.2 per 100 pages; Supplemental Table S3). Categories such as Access to Care (0.3 per 100 pages) and Emotional Support (0.3 per 100 pages) were the least frequently included. The distribution of recommendations

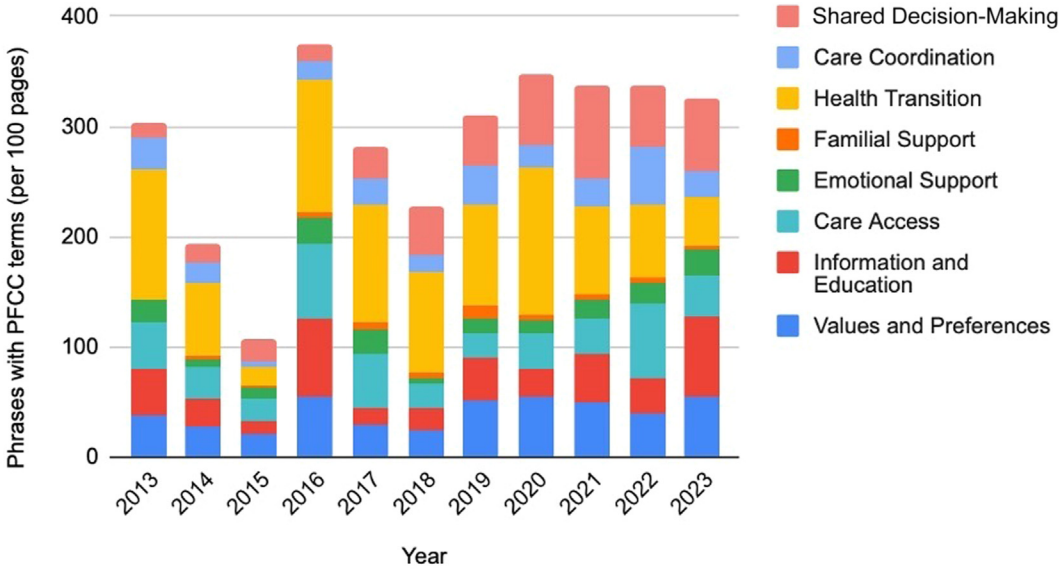

Figure 1. Trends in patient- and family-centred care (PFCC) dimensions over time, across societies (American College of Cardiology, American Heart Association, Canadian Cardiovascular Society, and European Society of Cardiology).). Shown are the yearly frequencies of PFCC dimensions across guidelines from 2013 to 2023. Frequencies reported are numbers of phrases containing PFCC terms per 100 pages.

and PFCC phrases across the PFCC dimensions among the societies are shown in Supplemental Fig. S4.

Trends in PFCC language usage over time

No increase occurred in overall PFCC usage over the study period ($P < 0.01$ for trend; Fig. 1). However, an increase in PFCC usage occurred for the ACC, AHA, and ESC (all $P < 0.001$ for trend), and a decrease occurred in PFCC usage for the CCS ($P = 0.003$ for trend) over the study period (Supplemental Fig. S5). When the CCS was excluded, all dimensions increased over the study period (all $P < 0.05$ for trend).

Topic-specific emphasis on PFCC terms

The most common topics for PFCC recommendations per 100 pages were heart failure (12.9), structural heart disease (11.6), valvular heart disease (11.2), and cardiovascular risk and disease (10.1; Supplemental Table S4). The least common topics for PFCC recommendations per 100 pages were vascular (8.0) and policy/training/education/methodology (8.9).

Discussion

Overview of PFCC-related language trends

This study investigated the prevalence and trends in PFCC-related language use and recommendations in major North American and European cardiovascular society guidelines.[11,12]

An increase occurred in PFCC language use over the past decade for the ACC, the AHA, and the ESC, but a decrease occurred for the CCS. The PFCC dimensions of Health Transition, SDM, and Care Access were the most frequently addressed overall, and Care Coordination, Emotional Support, and Familial Support were the least-emphasized dimensions. Notably, the ACC and AHA guidelines placed a greater emphasis on the dimensions of SDM and Values and Preferences, whereas the ESC guidelines more frequently underscored Health Transition and Information and Education. In contrast, the CCS guidelines highlighted Health Transition and Care Coordination as key areas.

The role of SDM in cardiovascular care

SDM was among the most frequently mentioned dimensions and key terms overall, and it was the most mentioned dimension and key term for the American societies. This finding aligns with the growing recognition of the importance of SDM in delivering high-quality, patient-centred cardiovascular care. SDM allows for the integration of clinical evidence with patient values and preferences in treatment decisions. SDM has been called the pinnacle of patient-centred care, particularly for preference-sensitive decisions for which multiple options are reasonable.[16] In addition, SDM is inextricably linked to evidence-based medicine, which is a foundational principle of cardiovascular medicine. SDM is essential for evidence-based medicine to achieve its aim, as it is the mechanism by which evidence can be brought explicitly into the consultation and discussed with the patient.[17] This relationship highlights the importance of including SDM recommendations in guidelines and providing practical guidance on how to effectively implement SDM in clinical practice.[18]

Patient values, health transitions, and underrepresented dimensions

The American societies' high-yield recommendations on respecting and adhering to patient values and preferences and health transitions within PFCC dimensions reflect their critical role in patient-centred cardiovascular care. Incorporating patient values into clinical decision-making helps tailor treatments to align with individual health goals, fostering better adherence and satisfaction with care.[19] This approach is particularly important in cardiovascular care, as it is an arena in which decisions often involve complex risk-benefit analyses.

Similarly, ensuring smooth health transitions—from hospital to home or across care settings—helps maintain continuity of care, reducing the risk of adverse events or hospital readmissions.[18] Together, these elements underscore the need for cardiology guidelines to offer practical guidance on how to integrate these critical components into routine care.

Care Coordination, Emotional Support, and Familial Support were the least frequently reported PFCC dimensions, yet these aspects are necessary for comprehensive patient care, particularly in the context of chronic cardiovascular conditions that require longitudinal management and support. These elements should be emphasized more robustly in future cardiology guidelines, to ensure comprehensive patient-centred care.

Care coordination is essential in cardiology, given the complexity of managing multiple care providers and transitions between healthcare settings. Effective care coordination ensures that treatment plans are followed across different stages of care, reducing the likelihood of miscommunication and adverse events.[20]

Moreover, emotional support is vital, as many cardiovascular patients face anxiety and depression, especially when managing chronic conditions such as heart failure.[21] Having emotional support can improve psychological well-being, which in turn influences adherence to medical recommendations and overall health outcomes.

The role of family involvement in cardiovascular care has been recognized increasingly, with the AHA's family-centred care statement highlighting the importance of engaging family members as a means to improve patient outcomes, particularly in complex decision-making and long-term disease management.[4,10] This recognition reflects the critical role families play in supporting adherence to treatment plans and providing emotional and practical support for patients.

Drivers of increased PFCC-related language usage over time

The decline in PFCC-related language usage within CCS guidelines over the study period contrasts with the upward trends observed in the ACC, AHA, and ESC guidelines, and it may reflect structural and contextual factors unique to the Canadian cardiovascular guideline development process. Unlike the ACC and AHA, which have invested in patient-facing initiatives—such as CardioSmart and the integration of patient representatives into guideline working groups—the CCS historically has emphasized clinical efficiency and evidence synthesis over patient-centred framing. Additionally, the CCS publishes shorter, more focused guidance documents, which could limit the breadth of language used to address the PFCC dimensions. The downward trend may also reflect shifts in publication priorities or a more conservative approach to language usage that emphasizes clinical utility over patient-centredness. Another possibility is that PFCC principles are embedded implicitly in CCS recommendations but are not articulated explicitly using standardized terminology, thus reducing their detectability by keyword-based analysis. Future efforts for the CCS could include collaboration with patient advisors, to ensure that PFCC principles are meaningfully represented and that PFCC is integrated into the CCS guideline development process.

PFCC in disease-specific guidelines

Heart failure, structural heart disease, and valvular disease guidelines had higher PFCC phrase counts compared to other topic areas, highlighting a recognition of PFCC's particular importance in these areas of cardiovascular medicine. Such emphasis may be due to the chronic nature of these conditions and the impact they have on patients' daily lives, necessitating a more patient-centred approach to management. For example, heart failure often involves complex symptom management, frequent hospitalizations, and decisions regarding advanced therapies, all of which require ongoing patient engagement and individualized care plans.[22] Similarly, structural heart disease management involves specific lifestyle elements, medication adherence, and lifelong monitoring, which may benefit particularly from shared decision-making to ensure that treatments align with patient preferences and long-term health goals.[23]

Challenges in implementing PFCC in practice

Despite the inclusion of many PFCC terms and recommendations in cardiovascular guidelines, the implementation of PFCC in routine cardiovascular care remains challenging. Research shows that organizational- and system-level characteristics play a crucial role in the failure to implement shared decision-making in routine care.[24] Our study suggests that professional societies should consider making patient-centred language a standard requirement when drafting guidelines, in a manner similar to how equity, diversity, and inclusion considerations are now emphasized. Guideline committees also could benefit from including patient and family partners, to ensure that the patient voice is represented adequately in the development process. Future research should focus on understanding how the incorporation of PFCC-related language influences healthcare professionals' practices and patient outcomes. Additionally, studies are needed to explore whether guidelines with stronger PFCC elements lead to improved patient engagement and adherence in cardiovascular care. These steps would help translate the increasing presence of PFCC-related language in guidelines into meaningful, practical changes in clinical practice.

Study strengths and limitations

This study has several strengths, including its comprehensive analysis of guidelines from major cardiovascular societies over the span of a decade, the use of advanced natural language processing techniques to identify PFCC language, and the categorization of findings into established PFCC dimensions. The study has several limitations. First, only 4 cardiovascular professional societies were included, which may not represent global trends in PFCC. However, the included societies are 4 of the largest and most influential cardiovascular societies. Second, the results were standardized to 100-page counts, to allow for direct comparisons between guidelines of varying lengths. This method, however, is not validated and may introduce bias in the interpretation of frequency data, especially given that the CCS guidelines are considerably shorter than the other societies' guidelines. Third, only the frequency of PFCC in-text citations and recommendations was analyzed and not the depth or quality of the recommendation. This study focused exclusively on quantifying the frequency of PFCC

recommendations and trends over time in cardiology guidelines. Although changes in the strength and content of these statements are equally important, they were outside the scope of our analysis. Future research should consider qualitative approaches to evaluate the depth and practical impact of PFCC recommendations on clinical practice and patient-centred outcomes. Such work would complement our findings and provide a more comprehensive understanding of how PFCC principles are integrated into cardiology guidelines. Additionally, we did not include executive summaries in our analysis. Executive summaries are often the primary resource for clinicians, and an analysis of these sections might yield different insights into PFCC use, as they often distill key points to make them easier to access.

Conclusion

An increasing trend was found in the inclusion of PFCC language and recommendations in cardiovascular guidelines for most cardiovascular societies. However, many PFCC dimensions, such as Care Coordination, Emotional Support, and Familial Involvement, remain underrepresented. By identifying these trends, this study provides a foundation for understanding how PFCC principles have been integrated into guidelines. Future research should evaluate how these recommendations are implemented in clinical practice and their influence on patient-centred outcomes, offering opportunities to further advance PFCC in cardiovascular care.

Data Availability

Any additional data can be made available upon request to the corresponding author.

Ethics Statement

This study is a systematic review of publicly available clinical guidelines and did not involve human participation or identifiable data.All research procedures adhered to the relevant ethical guidelines for research integrity and reporting.

Patient Consent

The authors confirm that patient consent is not applicable to this article. This study is a systematic review of publicly available clinical guidelines and did not involve any individual patient data.

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Disclosures

The authors have no conflicts of interest to disclose.

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Supplementary Material

To access the supplementary material accompanying this article, visit *CJC Open* at https://www.cjcopen.ca/ and at https://doi.org/10.1016/j.cjco.2025.05.017.